OR-AS Operations Research Applications and Solutions	Case Name: Retirement Apartments	Sector	Construction (residential building)
	OR-AS Operations Research - Applications and Solutions www.or-as.be info@or-as.be	Baseline Schedule	Schedule with resources Schedule with costs
		Risk Analysis	Random simulation One of nine std. scenarios User defined distributions
Submitted by	Torrino Debal		
Date	2015	Project Control	Automatic tracking Tracking based on user input
File Name	C2015-35 Retirement Apartments		

1. Project description

Project authenticity

The construction of retirement apartments, including structural works, installations, and finishing activities.

The project consists of activity, resource, and cost data obtained directly from the actual project execution.

2. Project properties

2.1. Baseline Schedule

General	
# Activities	13
Planned Duration (PD)	850d
Budget At Completion (BAC)	14,956,314.25€
Renewable Resources	-
Consumable Resources	-

Network topology	
Serial/Parallel (SP)	44%
Activity Distribution (AD)	75%
Length of Arcs (LA)	20%
Topological Float (TF)	30%

* standard eight-hour working days

2.2. Risk Analysis

Random simulation by ProTrack was performed using the default symmetric triangular risk distribution profiles.

	Cost sensitivity		
	avg [%]	std dev [%]	skew [-]
CRI-r	N/A	N/A	N/A
CRI-rho	N/A	N/A	N/A
CRI-tau	N/A	N/A	N/A

	Time sensitivity		
	avg [%]	std dev [%]	skew [-]
CI	N/A	N/A	N/A
SI	N/A	N/A	N/A
SSI	N/A	N/A	N/A
CRI-r	N/A	N/A	N/A
CRI-rho	N/A	N/A	N/A
CRI-tau	N/A	N/A	N/A

	Resource sensitivity		
	avg [%]	std dev [%]	skew [-]
CRI-r	N/A	N/A	N/A
CRI-rho	N/A	N/A	N/A
CRI-tau	N/A	N/A	N/A

2.3. Project Control

2.3.1. Simulated forecasting accuracy

The accuracy of time and cost forecasting methods has been evaluated based on Monte Carlo simulation runs using the risk profiles described in section “2.2. Risk Analysis”. Based on these risk profiles, the Mean Absolute Percentage Error (MAPE) and Mean Percentage Error (MPE) has been calculated to evaluate the expected accuracy of the time and cost predictions, EAC(t) and EAC, respectively.

Simulated EAC(t) accuracy		
method - PF	MAPE [%]	MPE [%]
PV - 1	N/A	N/A
PV - SPI	N/A	N/A
PV - SCI	N/A	N/A
ED - 1	N/A	N/A
ED - SPI	N/A	N/A
ED - SCI	N/A	N/A
ES - 1	N/A	N/A
ES - SPI(t)	N/A	N/A
ES - SCI(t)	N/A	N/A

Simulated EAC accuracy		
method (PF)	MAPE [%]	MPE [%]
1	N/A	N/A
CPI	N/A	N/A
SPI	N/A	N/A
SPI(t)	N/A	N/A
SCI	N/A	N/A
SCI(t)	N/A	N/A
0.8 CPI + 0.2 SPI	N/A	N/A
0.8 CPI + 0.2 SPI(t)	N/A	N/A

2.3.2. Tracking description

Tracking authenticity

Manual tracking was performed over-tracking periods. The Real Duration and Real Cost mentioned in section “2.3.3. Earned Value Management” are based on manual user input.

The tracking information obtained from the project owner and introduced in ProTrack includes actual activity start dates, durations and costs.

[1] The MAPE gives the best indication for the forecast accuracy (the lower the MAPE, the more accurate the method) since all deviations from the targeted real duration (real cost) are cumulated, whereas for the MPE underestimates can be compensated by overestimates and vice versa, possibly leading to an overly positive evaluation of a certain method. However, the MPE can provide useful information about the nature of the deviations, i.e. does the method rather underestimate or overestimate the real duration (real cost)?

2.3.3. Earned Value Management

2.3.3.1. Performance metrics

	CV [€]	SV [€]	SV(t) [d]	CPI [-]	SPI [-]	SPI(t) [-]	p-factor [-]
avg	N/A	N/A	N/A	N/A	N/A	N/A	N/A
std dev	N/A	N/A	N/A	N/A	N/A	N/A	N/A
final	N/A	N/A	N/A	N/A	N/A	N/A	N/A

2.3.3.2. Time forecasting

PD	850	Real Duration	951	Late/early	11%	

EAC(t)			Real Accuracy	
method - PF	avg [d]	std dev [d]	MAPE [%]	MPE [%]
PV - 1	N/A	N/A	N/A	N/A
PV - SPI	N/A	N/A	N/A	N/A
PV - SCI	N/A	N/A	N/A	N/A
ED - 1	N/A	N/A	N/A	N/A
ED - SPI	N/A	N/A	N/A	N/A
ED - SCI	N/A	N/A	N/A	N/A
ES - 1	N/A	N/A	N/A	N/A
ES - SPI(t)	N/A	N/A	N/A	N/A
ES - SCI(t)	N/A	N/A	N/A	N/A

2.3.3.3. Cost forecasting

BAC	14,956,314.25€	Real Cost	16,068,878.30€	Over/under budget	7.43%

EAC			Real Accuracy	
method (PF)	avg [€]	std dev [€]	MAPE [%]	MPE [%]
1	N/A	N/A	N/A	N/A
CPI	N/A	N/A	N/A	N/A
SPI	N/A	N/A	N/A	N/A
SPI(t)	N/A	N/A	N/A	N/A
SCI	N/A	N/A	N/A	N/A
SCI(t)	N/A	N/A	N/A	N/A
0.8 CPI + 0.2 SPI	N/A	N/A	N/A	N/A
0.8 CPI + 0.2 SPI(t)	N/A	N/A	N/A	N/A